

Python Fundamentals for Generative AI

Task 1: Python Basics & Operators

1. Create a Python script that accepts two numbers from the user.
2. Perform all basic arithmetic operations (add, subtract, multiply, divide, modulus, power).
3. Demonstrate use of comparison and logical operators (>, <, ==, !=, and, or, not).
4. Print results clearly using formatted output (f-strings).

Task 2: Lists and Arrays

1. Create a list of 10 random integers.
2. Perform the following operations:
 - Add and remove elements.
 - Find the maximum and minimum values.
 - Sort the list.
3. Convert the list into a NumPy array and calculate:
 - Mean, Median, and Standard Deviation.

Task 3: Dictionaries and Sets

1. Create a dictionary named student with keys: name, course, marks.
2. Add a new key grade and assign a value based on marks (e.g., A/B/C).
3. Print all keys and values using a loop.
4. Create two sets of AI tools,

Task 4: File Handling

1. Create a text file named ai_students.txt.
2. Write details of at least 5 students (name, marks, grade).
3. Read the file and display students who scored above 75 marks.

Task 5: Real-World Mini Project

Create a Python program named `ai_prompt_logger.py` that:

1. Accepts a user input (simulating an AI prompt).
2. Saves the prompt text in a file named `prompt_history.txt`.
3. Each time you run the program, new prompts are appended with timestamps.

Submission Instructions

1. Save your file as:
`YourName_GenAI_Week2.py`
2. Add comments explaining each code block.
3. Upload your completed script(s) to your GitHub repository under a folder named `Week2_Assignments`.
4. Share your GitHub link in the assignment Whatsapp Group.

DEADLINE : SUNDAY 11AM–2/11/2025